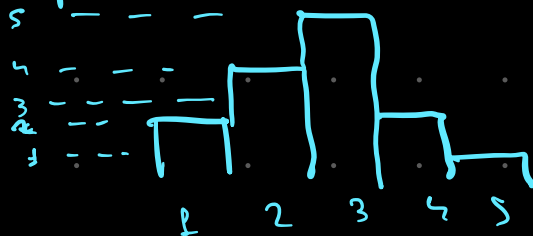


Noção de Probabilidade

o Probabilidade Discreta



Valores Inteiros

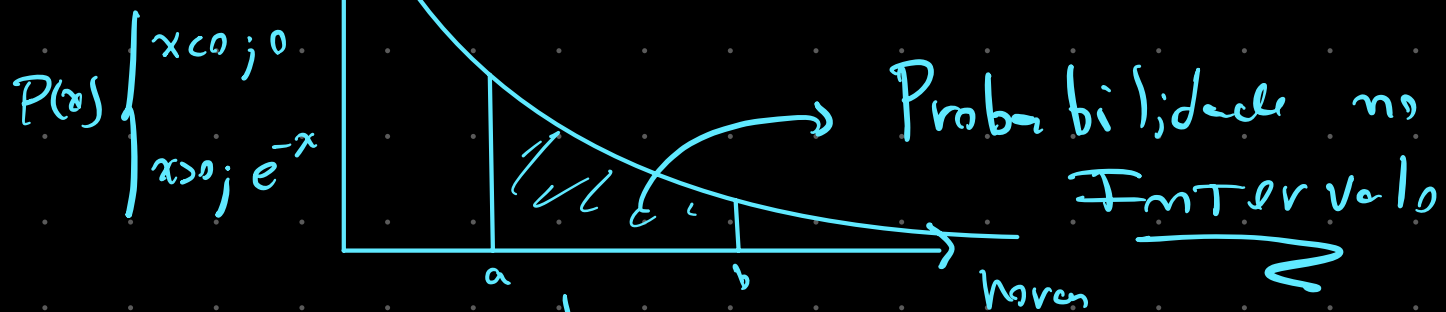
(1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 5)

1 : 2/15 ; 2 : 4/15 ; 3 : 5/15 ; 4 : 3/15 ; 5 : 1/15

o Probabilidade Contínua

$P(x)$

$P(x) \rightarrow$ Densidade de probabilidade



$$\text{Prob}(a \leq x \leq b) = \int_a^b P(x) dx$$

Prob $\rightarrow x = L$

$$\int_{-\infty}^{\infty} P(x) dx = 1$$

$$\text{Prob}(-\infty, \infty) = \int_{-\infty}^{\infty} P(x) dx = 1$$

Q evento mais provável

o Discreta

(0,0,1,1,1,2,3,3) mais provável = 1

Valor Médio (median)

o Discreta (0,1,1,1,2,3,3)

↓
1.5

o Contínua

$$p(x) = e^{-x}$$

$$\int_{-\infty}^{\text{median}} p(x) dx = 0.5$$

$$\int_{-\infty}^{\text{median}} e^{-x} dx = 0.5$$

$$-e^{-x} \Big|_0^{\text{median}} = 0.5$$

$$-e^{-\text{median}} + e^0 = 0.5$$

$$e^{-m} = 0.5$$

$$\ln 2 = m$$

• Média (mean μ)

• Discrete

(0, 0, 1, 1, 2, 2, 2, 3, 4, 5)

$$\frac{2 \cdot 0 + 3 \cdot 1 + 2 \cdot 2 + 1 \cdot 3 + 1 \cdot 4 + 1 \cdot 5}{10} = 0 \cdot \frac{2}{10} + 1 \cdot \frac{3}{10} + \dots$$

$$\mu = \sum_i x_i \cdot \text{Prob}(X=x_i) \quad \begin{matrix} \nearrow \\ 11 \\ 1.9 \end{matrix}$$

• Contínua

$$\mu = \int_{-\infty}^{\infty} x p(x) dx$$

para o caso e^{-x}

$$\mu = \int_0^{\infty} x e^{-x} dx$$

μ Também é chamado de
Valor esperado de uma função
(Expected value) $\langle x \rangle$

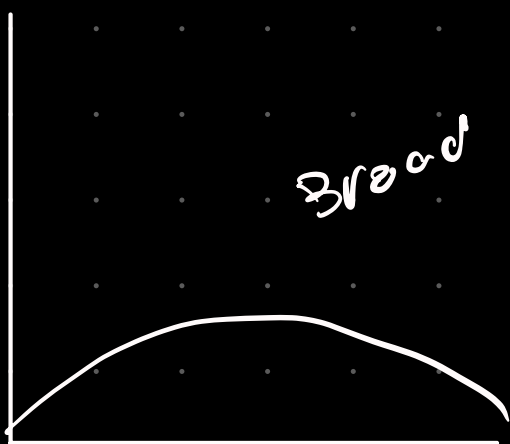
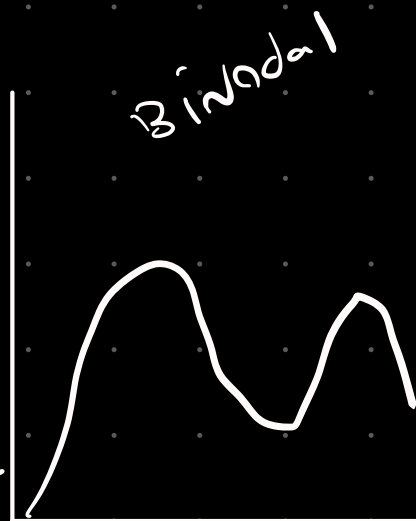
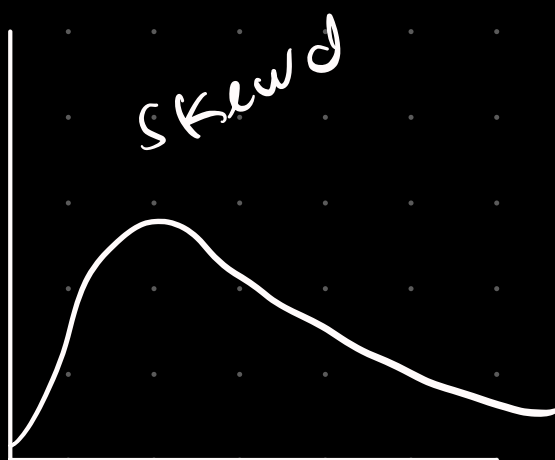
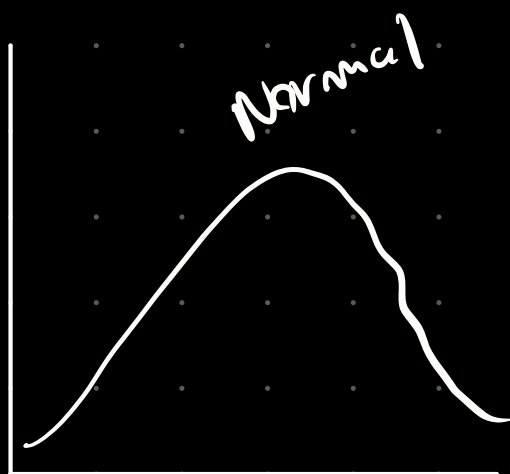
$$\rightarrow \int p(x) dx$$

$$\langle x \rangle = \sum_i f(x_i) \cdot \text{Prob}(x=x_i)$$

$$\langle x \rangle = \int_{-\infty}^{\infty} f(x) p(x) dx$$

Variancia e STD Deviation

Tipos de gráfico



Broadness (espessamento)

↓
o quão localizada
é a função

Variance $\rightarrow$ Broadness

$$\Delta x = x - \langle x \rangle \quad -1, 0, 1$$

$$\langle \Delta x \rangle = 0$$

$$\langle \Delta x^2 \rangle = \sigma^2 = \text{Variance} \quad 1, 0, 1$$

$\downarrow$ parameter significance + size

$$\sqrt{\frac{2}{3}}$$

RMS root mean square

$$\langle x^2 \rangle - \langle x \rangle^2 = \sigma^2$$

Example

(1, 2, 4, 5, 8)

x	$\Delta x = x - \langle x \rangle$	Δx^2	x^2
1	-3	9	1
2	-2	4	4
4	0	0	16
5	1	1	25
8	4	16	64

$$\langle x \rangle = 4 \quad \text{~~~~~} \quad \langle \Delta x^2 \rangle \quad \langle x \rangle^2 = 16$$

$\text{''} = \sigma^2$
6

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 \Rightarrow$$

$$\langle \Delta x^2 \rangle = 6 \quad 22 - 16 = 6 \quad \text{var}$$

Previsão Intervalos a partir de $\langle \rangle$

$$\langle x \rangle = \int x p(x) dx \rightarrow \mu$$

$$\langle x^2 \rangle = \int x^2 p(x) dx \rightarrow \propto \sigma^2$$

$$\langle x^3 \rangle = \int x^3 p(x) dx \rightarrow \propto \text{Skewness}$$

$$\langle x^4 \rangle = \int x^4 p(x) dx \rightarrow \propto \text{Broadness}$$

Normalização de Probabilidade

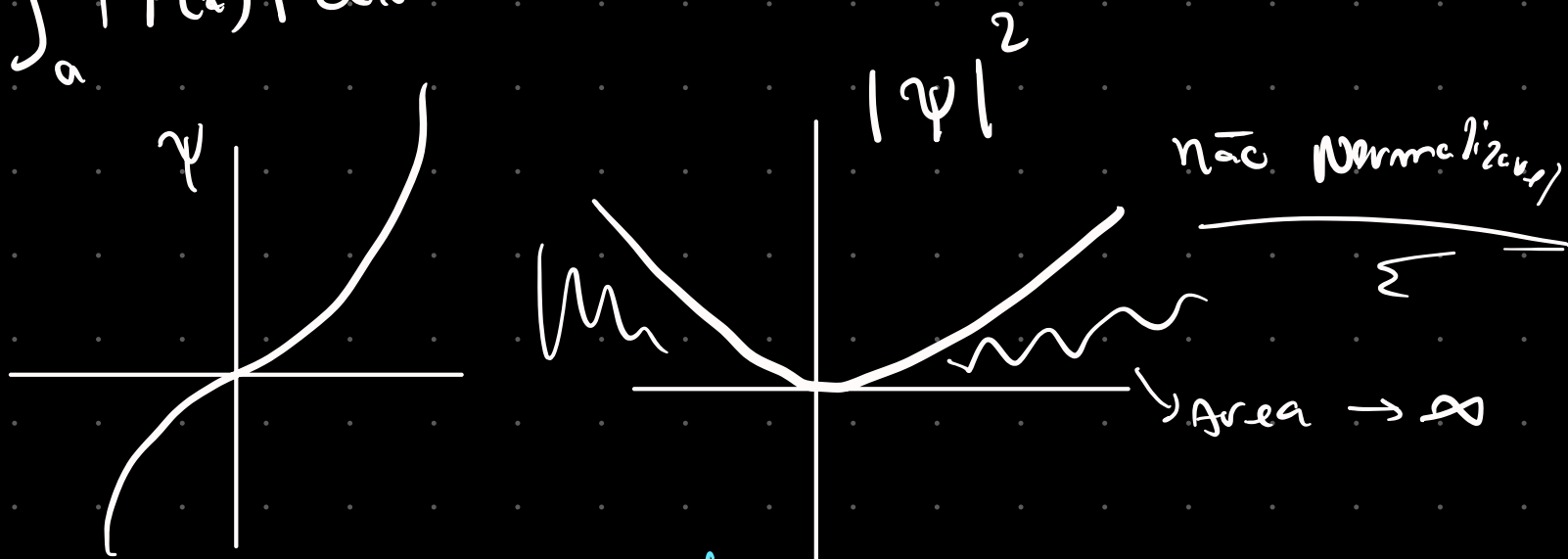
→ Para ser solução válida da Eq Sch.

ψ tem que ser viável na probabilidade

ou seja $\int_{-\infty}^{\infty} p(x) dx = 1$

$$P(x) = |\psi(x)|^2$$

$$\int_a^b |\psi(x)|^2 dx = 1$$



Para Ser Normalizável

$$\psi \rightarrow 0 \text{ quando } x \rightarrow \pm \infty$$

$$\psi(x, \tau) = A e^{\frac{i\hbar}{\hbar} (px - Et)}$$

Normalização e a evolução
no Tempo

$$|\psi(x)|^2 = \psi \psi^*$$

$$\frac{d}{d\tau} \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial \tau} [\psi \psi^*] dx$$

$$\int_{-\infty}^{\infty} \left\{ \psi \frac{\partial \psi^*}{\partial \tau} + \psi^* \frac{\partial \psi}{\partial \tau} \right\} dx$$

Usando Eq. SCH pro deixar todos valores de Integrais em função de x

$$i\hbar \frac{\partial \psi}{\partial \tau} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi$$

$$\frac{\partial \psi}{\partial \tau} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} \cdot \frac{i}{\hbar} + V(x) \psi \cdot \frac{i}{\hbar}$$

$$\frac{\partial \psi}{\partial \tau} = \frac{i\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - V(x) \psi \frac{i}{\hbar}$$

$$\frac{\partial \psi^*}{\partial \tau} = -\frac{i}{2m} \hbar^2 \frac{\partial^2 \psi^*}{\partial x^2} + V(x) \psi^* \frac{i}{\hbar}$$

$$\int_{-\infty}^{\infty} \left[\psi \left(-\frac{i}{2m} \hbar^2 \frac{\partial^2 \psi^*}{\partial x^2} + V(x) \psi^* \frac{i}{\hbar} \right) + \psi^* \left(\frac{i\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - V(x) \psi \frac{i}{\hbar} \right) \right] dx$$

$$\forall V(x) = 0$$

$$\int_{-\infty}^{\infty} \frac{i\hbar^2}{2m} \left[-\psi \frac{\partial^2 \psi^*}{\partial x^2} + \psi^* \frac{\partial^2 \psi}{\partial x^2} \right] dx$$

$$\frac{i\hbar^2}{2m} \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right] dx$$

$$\frac{i\hbar^2}{2m} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) \Big|_{-\infty}^{\infty} = Cte$$

Para ser Normalizável

$$x \rightarrow \pm\infty \quad \psi \rightarrow 0$$

se ψ existe

Variações
no tempo
não implicam

na Normalização

derivadas parciais
só de x

Só depende
do ponto
Espacial

e não do tempo.

Ex $\psi(x) \begin{cases} 0 & ; |x| > 1 \\ A e^{ix}(1-x^2) & \text{for } -1 \leq x \leq 1 \end{cases}$

$$1 = \int_{-1}^1 A e^{ix}(1-x^2) \cdot A e^{-ix}(1-x^2) dx$$

$$1 = A^2 \int_{-1}^1 (1-x^2) e^{ix} \cdot e^{-ix} dx$$

$$1 = A^2 \int_{-1}^1 (1-x^2) dx$$

$$1 = A^2 \int_{-1}^1 (1 - 2x^2 + x^4) dx$$

$$1 = A^2 \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right] \Big|_{-1}^1$$

$$1 = A^2 \left[1 - \frac{2}{3} + \frac{1}{5} + 1 - \frac{2}{3} + \frac{1}{5} \right]$$

$$A^2 = \frac{15}{16}$$

$$\left(A = \frac{\sqrt{15}}{4} \right) \text{ C.T. } \text{do Normalizaco}$$

Operadores na Quântica

Operadores representam medição na Quântica assim aplicar um operador muda a função de onda

medir movimento de uma partícula não é possível já que medir o momento ou posição em um ponto muda o comportamento da mesma

Para assim fazer a medição é necessário ter varias cópias da partícula e medir suas posições em instantes diferentes

$$Prob = \int_a^b \psi \psi^* dx$$

$$\langle x \rangle = \int_a^b \overbrace{x \psi \psi^*}^{\text{operador}} dx$$



valor esperado do operador de posição

Determinando $\langle V \rangle$ do operador $\hat{V}$

$$\frac{d\langle x \rangle}{d\tau} = \frac{d}{d\tau} \int_a^b x \psi \psi^* dx$$

$$\int_a^b x \frac{\partial (\psi \psi^*)}{\partial \tau} dx$$

$$\frac{d\langle x \rangle}{d\tau} = \frac{i\hbar}{2m} \int_a^b x \frac{\partial}{\partial \tau} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) dx$$

$$\langle \hat{V} \rangle = -\frac{i\hbar}{m} \int \psi^* \frac{\partial \psi}{\partial x} dx$$

Valor esperado de Operadores

$$\langle \hat{V} \rangle = -\frac{i\hbar}{m} \int \psi^* \frac{\partial}{\partial x} \psi dx$$

$$\langle \hat{P} \rangle = -i\hbar \int \psi^* \frac{\partial}{\partial x} \psi dx$$

$$\langle \hat{x} \rangle = \int \psi^*(x) x \psi dx$$

$$\text{no Geral } \langle \hat{A} \rangle = \int \psi^* \hat{A} \psi dx$$

O perador

Posição $\hat{x} = x \cdot \square$

Velocidade $\hat{v} = \frac{-i\hbar}{m} \frac{\partial}{\partial x} \square$

Momento $\hat{p} = -i\hbar \frac{\partial}{\partial x} \square$

Energia cinética $\hat{T} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \square$

A equação de Schrödinger

$$\hat{H}\psi = E\psi$$

a equação é uma característica fundamental assim como a conservação de energia
então ela não é "derivada" mas sim deduzida

a Eq Sch é construída de forma
que ψ seja sua solução
então é construída a partir do ψ
Para relacionar Momento e Energia
é conhecido $p = mv$ $E = \frac{mv^2}{2}$

$$E = \frac{p^2}{2m}$$

a função de onda

$$\psi(x, \tau) = A e^{i(kx - \omega\tau)}$$

$$k = \frac{p}{\hbar} \quad \omega = \frac{E}{\hbar}$$

$$\psi(x, \tau) = A e^{\frac{i}{\hbar}(px - Et)}$$

$$\frac{\partial \psi}{\partial x} = \frac{i}{\hbar} p \psi$$

$$\frac{\partial \psi}{\partial \tau} = -\frac{i}{\hbar} E \psi$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi$$

$$E = \frac{p^2}{2m}$$

$$E\psi = \frac{p^2}{2m} \psi$$

$$p^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

$$E\psi = i\hbar \frac{\partial \psi}{\partial t}$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} \text{ pode ser}$$

$$\hat{K} \psi = E \psi \quad \text{lev como o operador } \hat{K} \text{ que age sobre } \psi$$

O problema $\rightarrow$ agem como o ato de observar na mecânica quântica

$$\hat{p} \hat{x} \psi \neq \hat{x} \hat{p} \psi$$

$$(\hat{K} + \hat{V}) \psi = E \psi$$

$$\hat{H} \psi = E \psi$$

↓
operator

↓
auto value
(eigen value)

Example

Determiner $\langle \hat{P} \rangle$ de $\psi \begin{cases} 0; |x| \geq 1 \\ A(1-x^2) \end{cases}$

$$\langle \hat{P} \rangle = \int_{-1}^1 A(1-x^2) \cdot \hat{P} A(1-x^2)$$

$$\langle \hat{P} \rangle = \int_{-1}^1 A(1-x^2) \cdot \frac{-i\hbar \partial}{\partial x} A(1-x^2) dx$$

$$\langle \hat{P} \rangle = \int_{-1}^1 -A(1-x^2) \cdot i\hbar A \cdot (-2x) dx$$

$$\langle \hat{P} \rangle = A^2 i\hbar \int_{-1}^1 (2x - 2x^3) dx$$

$$\langle \hat{P} \rangle = A^2 i\hbar \left(x^2 - \frac{1}{4} x^4 \right) \Big|_{-1}^1$$

$$\langle \hat{p} \rangle = A^2 i \hbar \left[\left(1 - \frac{1}{4}\right) - \left(1 - \frac{1}{4}\right) \right]$$

$\langle \hat{p} \rangle = 0 \rightarrow$ é uma onda simétrica
da não aparenta tendência de movimento

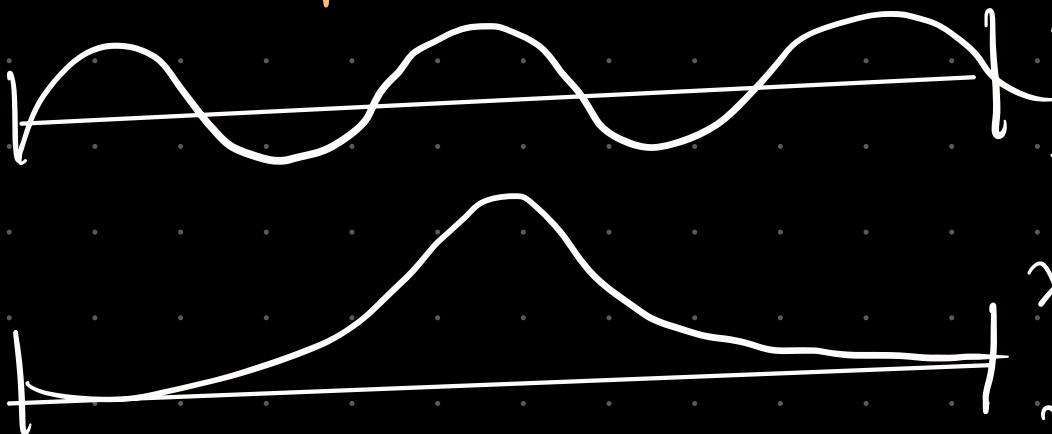


na média
não se move

Importante $\langle \hat{Q} \rangle = \int \psi^* \hat{Q} \psi dx$ ✓
 $\int \hat{Q} \psi^* \psi dx$ ✗

Ja que $\hat{Q}$ esta
atuando em ψ

Princípio Da Incerteza



λ conhecido
Preciso

x deslocalizado
Não sabe onde

λ desconhecido
Incerto

x localizado
conhecido

$$\Delta \lambda \Delta x \geq 1$$

a incerteza
de λ multiplicado
pelo Incerteza de x

Prova por
análise de Fourier que 1

Tem que ser maior

De Broglie particula onda

$$E = mc^2 \quad E = hf \quad f = \frac{c}{\lambda}$$

$$hf = mc^2$$

$$\frac{h}{\lambda} = mc$$

$$\frac{h}{\lambda} = \underline{1mc}$$

Atomo de
de Broglie

$$p = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{p}$$

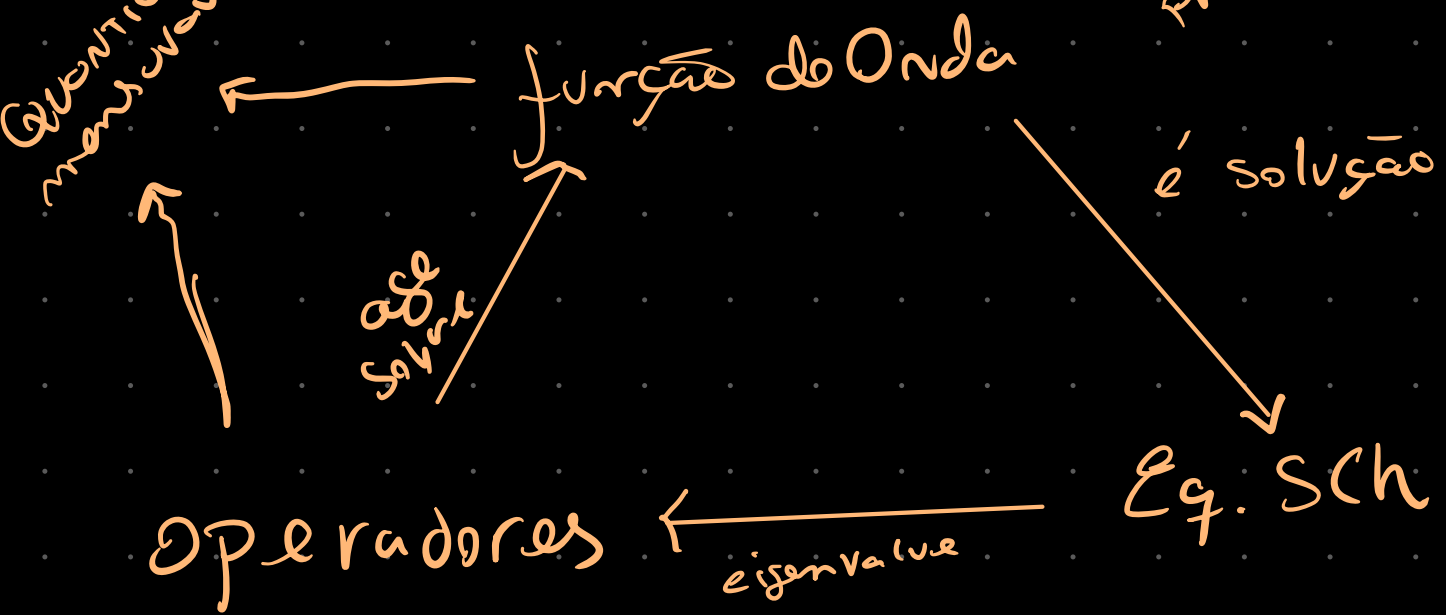


$$\Delta p \cdot \Delta x \geq \frac{h}{2}$$

$$\Delta E \Delta \tau \geq \frac{h}{2}$$

ident

→ Incerteza
de
probabilidade



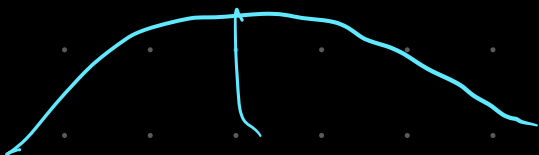
densidade de probabilidade $P(x) = |\psi(x)|^2$

Probabilidade $\int P(x) dx$

Normalizar $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$

Variancia $\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \int x^2 P(x) dx$

↙
e qualquer operador



$\langle \bar{x} \rangle$



ODE and PDE
(EDO) (EDP)

$x(z), y(z), v_x(z), v_y(z)$

$$\frac{dx}{dz} = v_x \quad \frac{dy}{dz} = v_y$$

$$\frac{dv_x}{dz} = -k v_x \quad \frac{dv_y}{dz} = -k v_y - \sigma$$

$\vec{E}(x, y, z)$

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{q_{int}}{\epsilon} \Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{q_{int}}{\epsilon}$$

$$\vec{E} = -\vec{\nabla} V \rightarrow \vec{\nabla}^2 V = \frac{q_{int}}{\epsilon}$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \frac{q_{int}}{\epsilon}$$

Separação de Variável

$$V(x, y, z) \Rightarrow X(x) \cdot Y(y) \cdot T(z)$$

Em Eq. da Onda Aceleração & Curvatura

$$\underbrace{\frac{\partial^2 \psi(x, z)}{\partial \tau^2}}_{\text{acceleration}} = \kappa \cdot \underbrace{\frac{\partial^2 \psi(x, z)}{\partial x^2}}_{\text{curvature}} \quad \left\{ \kappa = c^2 \right\}$$

$$\frac{\partial^2 [\chi(x) \tau(\tau)]}{\partial \tau^2} = c^2 \frac{\partial^2 [\chi(x) \tau(\tau)]}{\partial x^2}$$

$$\frac{\partial}{\partial \tau} \left[\chi(x) \frac{\partial \tau(\tau)}{\partial \tau} + \tau(\tau) \frac{\partial \chi(x)}{\partial \tau} \right] \quad \nabla^2 (r, \theta, \phi)$$

$$\chi(x) \frac{\partial^2 \tau}{\partial \tau^2} + \frac{\partial \tau}{\partial \tau} \frac{\partial \chi}{\partial \tau} + \tau \frac{\partial^2 \chi}{\partial \tau^2} + \frac{\partial \chi}{\partial \tau} \frac{\partial \tau}{\partial \tau}$$

$$= c^2 \left[\frac{\partial^2 \tau}{\partial x^2} \left(\frac{\partial \tau}{\partial x} \frac{\partial \chi}{\partial x} \right) + \frac{\partial^2 \chi}{\partial x^2} \right]$$

$$\chi(x) \frac{\partial^2 \tau}{\partial \tau^2} = c^2 \tau(\tau) \frac{\partial^2 \chi}{\partial x^2} \quad \left. \begin{array}{l} \text{EDP} \\ \text{EDO} \end{array} \right\}$$

$$\frac{1}{\tau} \frac{\partial^2 \tau}{\partial \tau^2} = \frac{c^2}{\chi} \frac{\partial^2 \chi}{\partial x^2}$$

depende
só de τ

depende só
de x

$$\text{Se } \tau(\tau) = x(x) = a$$

$$\frac{1}{T} \frac{d^2 \tau}{d\tau^2} = a$$

$$\frac{c^2}{x} \frac{d^2 x}{dx^2} = a$$

$$\frac{d^2 \tau}{d\tau^2} = a\tau$$

$$\frac{d^2 x}{dx^2} = \frac{a}{c^2} x$$

$$\tau = e^{\sqrt{a} \tau}$$

$$x = e^{\frac{\sqrt{a}}{c} x}$$

$$\psi(x, \tau) = \tau(\tau) \cdot x(x)$$

$$\psi(x, \tau) = A e^{\sqrt{a} \left(\tau + \frac{x}{c} \right)}$$

A determined by boundary Conditions

$a \rightarrow$ complexo normalizado

Exercício "The Heat Equation"

$$\frac{\partial u(x, \tau)}{\partial \tau} = \alpha \frac{\partial^2 u(x, \tau)}{\partial x^2}$$

$$\frac{\partial (x)(\tau)}{\partial \tau} = \alpha \frac{\partial^2 x \cdot \tau}{\partial x^2}$$

$$x \frac{\partial \tau}{\partial \tau} = a \tau \frac{\partial^2 x}{\partial x^2}$$

$$\frac{1}{\tau} \frac{\partial \tau}{\partial \tau} = \frac{a}{x} \frac{\partial^2 x}{\partial x^2} \Rightarrow \frac{1}{\tau} \frac{d\tau}{d\tau} = \frac{a}{x} \frac{d^2 x}{dx^2}$$

$$\frac{d\tau}{d\tau} = a\tau$$

$$\frac{\partial^2 x}{\partial x^2} = \frac{ax}{x}$$

$$\sqrt{\frac{a}{x}} x$$

$$T(\tau) = e^{a\tau} ; x(x) = e$$

$$U(x, \tau) \Rightarrow A e^{a\tau} \cdot e^{\sqrt{\frac{a}{x}} x}$$

$$U(x, \tau) = A e^{a\tau + \sqrt{\frac{a}{x}} x}$$

$A \rightarrow$ Determinado por condição limite

$a \rightarrow$ Complexo

A Eq de Schrödinger Independente do Tempo (TISE)

$$i\hbar \frac{\partial \psi}{\partial \tau} = \hat{H} \psi$$

$$i\hbar \frac{\partial \psi(x, \tau)}{\partial \tau} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, \tau)}{\partial x^2} + V(x) \psi(x, \tau)$$

$$i\hbar \frac{\partial [x, \tau]}{\partial \tau}$$

$$i\hbar x \frac{\partial \tau}{\partial \tau} = -\frac{\hbar^2}{2m} \tau \frac{\partial^2 x}{\partial x^2}$$

$$\frac{i\hbar}{\tau} \frac{\partial \tau}{\partial \tau} = \frac{-\hbar^2}{2m x} \frac{\partial^2 x}{\partial x^2} = \mathcal{E}$$

Independente
do tempo

Dependente do Espaço

$$i\hbar \frac{\partial \tau}{\partial \tau} = \tau \mathcal{E}$$

$$\frac{\partial \tau}{\partial \tau} = -\frac{i}{\hbar} \tau E$$

$$\tau = A e^{-\frac{i}{\hbar} E \tau}$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \chi}{\partial x^2} = E \chi$$

$$\hat{H} \chi = E \chi$$

$$\Psi(x, \tau) = \Psi(x) \cdot \Psi(\tau)$$

Eq Sch
Independent do Time

Soluções Estacionárias
Para Eq. Sch (TISE)

$$\hat{H} \Psi = E \Psi$$

conceitos

$$\Psi_s(x, \tau)$$

$$\Psi_s(x, \tau) = \Psi(x) \cdot e^{-\frac{i}{\hbar} E \tau} \quad \Psi(x)$$

$$P(x) = |\Psi_s|^2 = \Psi_s^* \Psi_s$$

$$P(x) = \Psi \cdot e^{-\frac{i}{\hbar} E \tau} \cdot \Psi^* e^{\frac{i}{\hbar} E \tau}$$

$$P(x) = \Psi \Psi^*$$

assim quando escrevemos
 $P(x) = \Psi_s \Psi_s^*$ e não permite
sistemas estacionários

$$\langle \hat{A} \rangle = \int \Psi_s^* \hat{A} \Psi_s dx$$

(que não evolui com o tempo)

$$\langle \hat{A} \rangle = \int \Psi^* e^{\frac{iEt}{\hbar}} \cdot \hat{A} \cdot \Psi \cdot e^{-\frac{iEt}{\hbar}} dx$$

$$\langle \hat{A} \rangle = \int \Psi^* \hat{A} \Psi dx$$

Se o operador $\hat{A}$ não for derivado Temporal (Independente no T)

Estados Estacionários

Energia (Ψ)

Usando do TISE $\hat{H}\Psi = E\Psi$

$$\langle \hat{H} \rangle = \int \Psi^* \hat{H} \Psi dx$$

$$\langle \hat{H} \rangle = \int \Psi^* E \Psi dx$$

$$\langle \hat{H} \rangle = E$$

$$\sigma^2 = \underbrace{\langle \hat{H}^2 \rangle} - \underbrace{\langle \hat{H} \rangle^2}_{E^2}$$

$$\langle \hat{H}^2 \rangle = \int \psi^* \hat{H} \cdot \hat{H} \psi \, dx$$

$$\langle \hat{H}^2 \rangle = \int \psi^* \hat{H} E \, dx$$

$$\langle \hat{H}^2 \rangle = \int \hat{H} \psi^* E \, dx$$

$$\langle \hat{H}^2 \rangle = E^2 \quad \text{Variancia de } \hat{H} \text{ Para um estado estacionário}$$

$$\sigma_{\hat{H}}^2 = 0$$

Sempre é 0

assim pelo princípio de Incerteza

$$\underbrace{\Delta E}_{0} \Delta \tau \geq \frac{\hbar}{2}$$

não há incerteza de energia

$$\Delta \tau \rightarrow \infty$$

é impossível saber quando um estado estacionário ocorre, é como se ele sempre existisse, é cte pra sempre $\rightarrow$ que não é realístico mas tem boas propriedades matemáticas

Superposição de Estados Estacionários

Eq. Sch é linear então se A e B são soluções $\hat{H} C = nA + mB$ tbm é

Princ Simplem

$$i\hbar \frac{\partial (a+b)}{\partial \tau} = \hat{H} (a+b)$$

$$i\hbar \frac{\partial a}{\partial \tau} + i\hbar \frac{\partial b}{\partial \tau} = \hat{H} a + \hat{H} b$$

Devido à isso podemos escrever

Solução Geral da Eq. Sch

$$\Psi(x, \tau) = \sum_{j=0}^n C_j \psi(x)_j \cdot e^{-\frac{i E_j \tau}{\hbar}}$$

Um estado estacionário / onda estacionária é a $\left[\psi(x) \cdot e^{-\frac{i E \tau}{\hbar}} \right]$

ela é dada tal que seja Solução
para TISE estado estacionário

$$\Psi(x, \tau) = \sum_{j=0}^n C_j \psi(x)_j \cdot e^{-\frac{i E_j \tau}{\hbar}}$$

uma onda é um somatório de estados
estacionários com fases diferentes

Superposing stationary states Ψ_{s1} Ψ_{s2}

$$\circ \hat{H} \Psi_{s1} = E \Psi_{s1}$$

$$\circ \hat{H} \Psi_{s2} = E \Psi_{s2}$$

$$\Psi(x, \tau=0) = C_1 \Psi_{s1} + C_2 \Psi_{s2}$$

$$\Psi(x, \tau) = C_1 \Psi_{s1} \cdot e^{-\frac{i E_1 \tau}{\hbar}} + C_2 \Psi_{s2} \cdot e^{-\frac{i E_2 \tau}{\hbar}}$$

Resolvendo a Energia para $\Psi(x, \tau)$

* uma onda não estacionária

$$\langle \hat{H} \rangle = \int \Psi^* \hat{H} \Psi \Rightarrow \int (C_1 \Psi_{s1}^* e^{\frac{i E_1 \tau}{\hbar}} + C_2 \Psi_{s2}^* e^{\frac{i E_2 \tau}{\hbar}}) \cdot \hat{H} (C_1 \Psi_{s1} e^{-\frac{i E_1 \tau}{\hbar}} + C_2 \Psi_{s2} e^{-\frac{i E_2 \tau}{\hbar}}) dx$$

Observe que o Hamiltoniano $\frac{-\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$ não faz nada com a Dependência Temporal

$$\langle \hat{H} \rangle = \int (C_1 \Psi_{s1}^* e^{\frac{i E_1 \tau}{\hbar}} + C_2 \Psi_{s2}^* e^{\frac{i E_2 \tau}{\hbar}}) \cdot \left[\underbrace{C_1 \hat{H} \Psi_{s1}}_{E_1 \Psi_{s1}} e^{-\frac{i E_1 \tau}{\hbar}} + C_2 \underbrace{\hat{H} \Psi_{s2}}_{E_2 \Psi_{s2}} e^{-\frac{i E_2 \tau}{\hbar}} \right] dx$$

$$\langle \hat{H} \rangle = \int (C_1 \psi_{s_1}^* e^{\frac{iE_1\tau}{\hbar}} + C_2 \psi_{s_2}^* e^{\frac{iE_2\tau}{\hbar}}) \cdot (C_1 \psi_{s_1} e^{-\frac{iE_1\tau}{\hbar}} + C_2 \psi_{s_2} e^{-\frac{iE_2\tau}{\hbar}}) dx$$

$$\alpha = \frac{iE_1\tau}{\hbar} \quad \beta = \frac{iE_2\tau}{\hbar}$$

$$\langle \hat{H} \rangle = \int (C_1 \psi_{s_1}^* e^{\alpha} + C_2 \psi_{s_2}^* e^{\beta}) (C_1 \psi_{s_1} e^{-\alpha} + C_2 \psi_{s_2} e^{-\beta}) dx$$

$$\int [C_1^2 \underbrace{\psi_{s_1}^*}_{\psi_{s_1}} \underbrace{\psi_{s_1}}_{\psi_{s_1}} + C_1 C_2 \underbrace{\psi_{s_1}^*}_{\psi_{s_1}} \underbrace{\psi_{s_2}}_{\psi_{s_2}} e^{\alpha-\beta} + C_1 C_2 \underbrace{\psi_{s_2}^*}_{\psi_{s_2}} \underbrace{\psi_{s_1}}_{\psi_{s_1}} e^{\beta-\alpha} + C_2^2 \underbrace{\psi_{s_2}^*}_{\psi_{s_2}} \underbrace{\psi_{s_2}}_{\psi_{s_2}}] dx$$

uma propriedade importante de Estados Estacionários
é a sua ortogonalidade $\int \psi_{s_1} \psi_{s_2}^* dx = 0$

$$\int [C_1^2 \psi_{s_1}^* \psi_{s_1} + C_2^2 \psi_{s_2}^* \psi_{s_2}] dx$$

$$\int C_1^2 \psi_{s_1}^* \psi_{s_1} + C_2^2 \psi_{s_2}^* \psi_{s_2} \Rightarrow C_1^2 \epsilon_1 + C_2^2 \epsilon_2 = \langle \hat{H} \rangle$$

→ Prova

$$\hat{H} \psi_1 = \epsilon_1 \psi_1 ; \langle \psi_2 | \hat{H} | \psi_1 \rangle = \langle \hat{H} \rangle \text{ Notação de Dirac}$$

$$\hat{H} \psi_2 = \epsilon_2 \psi_2 ; \langle \psi_2 | \hat{H} | \psi_1 \rangle = \epsilon_1 \langle \psi_2 | \psi_1 \rangle = \epsilon_2 \langle \psi_2 | \psi_1 \rangle$$

$$0 = \langle \psi_2 | \psi_1 \rangle (\epsilon_1 - \epsilon_2)$$

Resolução Quântica Clássica

- 1) Começar com um Sistema físico
- 2) anotar o potencial $V(x)$
- 3) Resolver TISE para esse sistema
(obter ψ)
- 4) Determinar os ESTADOS ESTACIONÁRIOS
do sistema $\Psi_s(x, \tau) = \psi(x) e^{-\frac{i E \tau}{\hbar}}$
- 5) Expressar a Condição Inicial
como a $\sum$ dos estados estacionários
no espaço de momento

$$\Psi(x, \tau=0) = \sum_j^n C_j \psi(x)$$

- 6) adicionar Dependência Temporal
a cada Estado Estacionário

$$\Psi(x, \tau) = \sum_j^n C_j \Psi_s(x, \tau)$$

$$\Psi(x, \tau) = \sum_j^n C_j \psi(x) e^{-\frac{i E_j \tau}{\hbar}}$$

That's it! Hope you like it!

funções Potenciais $V(x)$ na Eq. Sch

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi \quad \text{TDE}$$

$$E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi \quad \text{TIE}$$

Schr

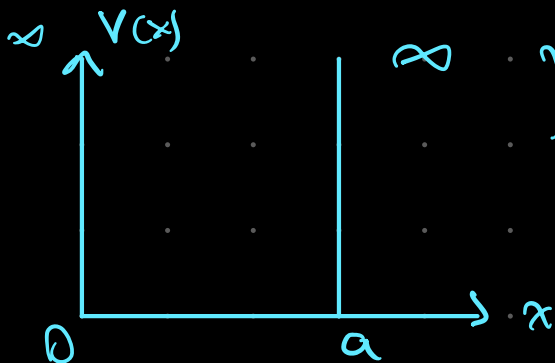
Energy of the
stationary state

$$-\frac{2m}{\hbar^2} [E - V(x)] \psi = \frac{\partial^2 \psi}{\partial x^2}$$

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{2m}{\hbar^2} [V(x) - E] \psi$$

$V(x)$

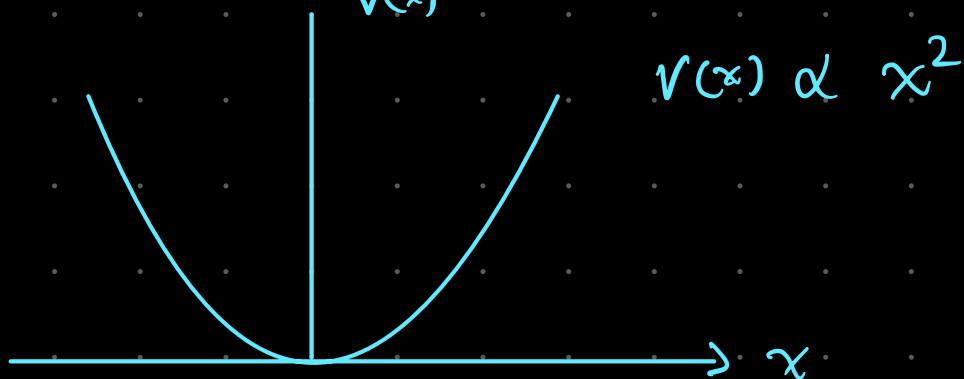
Particle in a Box // Infinite Square Well



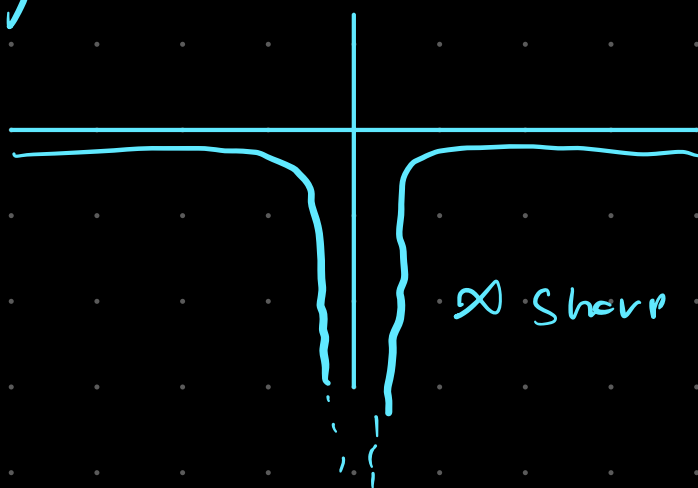
$$V(x > a \vee x < 0) = \infty$$

$$V(a > x > 0) = 0$$

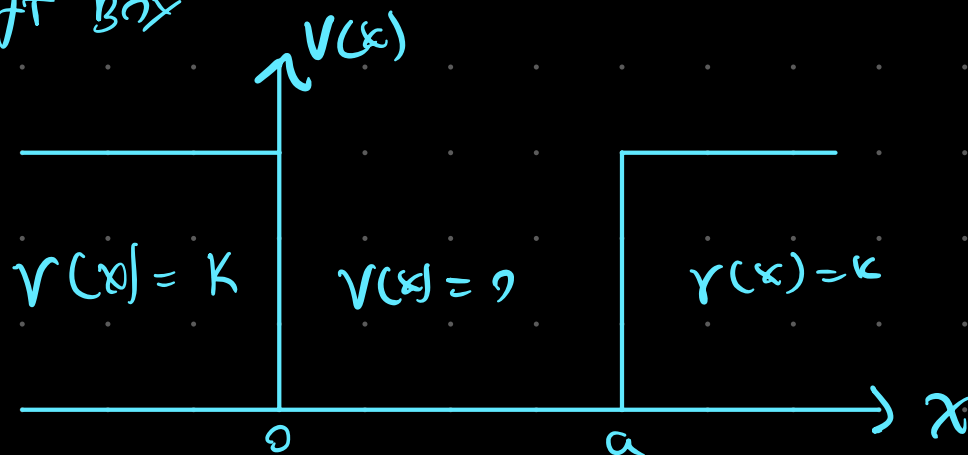
Harmonic Oscillator



δ function potential



Particle in a soft box



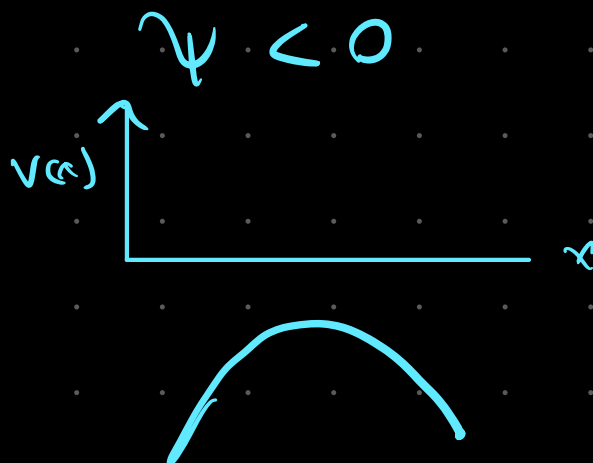
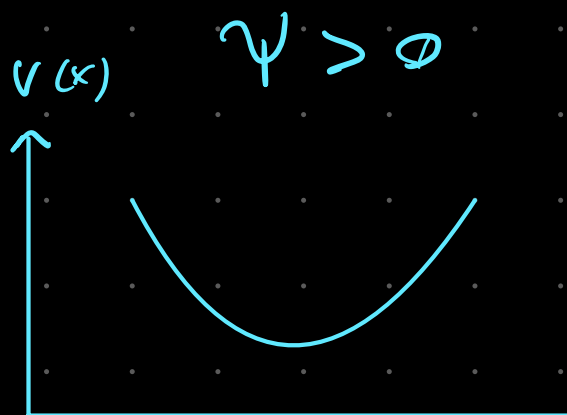
Physical Significance of

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{2m}{\hbar^2} [V(x) - E] \psi$$

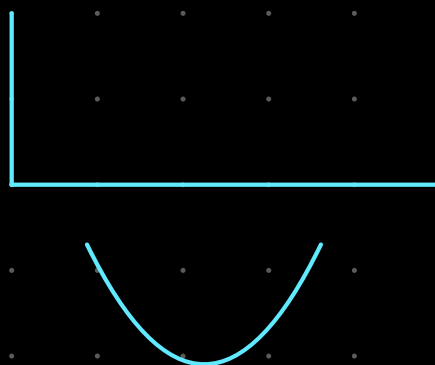
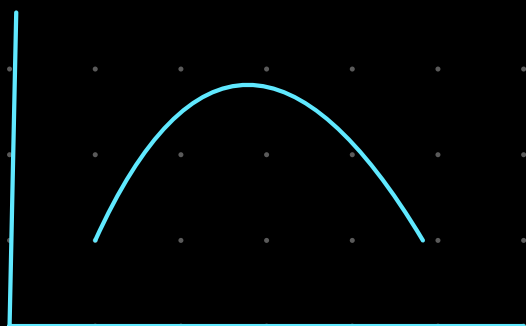
→ $\frac{\partial \psi}{\partial x}$ → Curvatura de ψ

→ $\frac{\partial^2 \psi}{\partial x^2}$ → Sentido da Curva $\uparrow$
 $\downarrow$

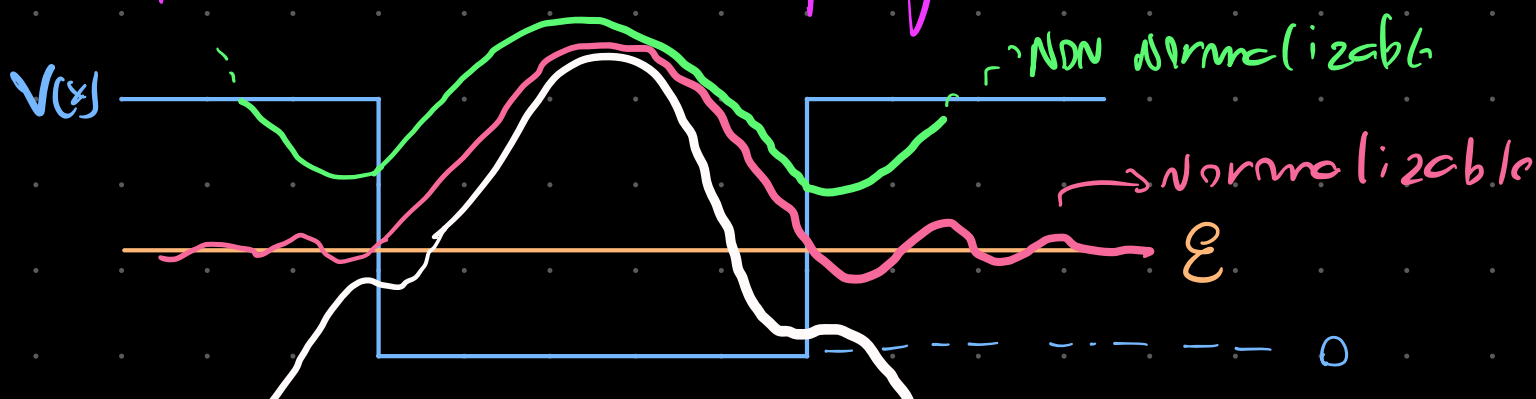
Se $V(x) > E$



Se $V(x) < E$
 $\psi > 0$



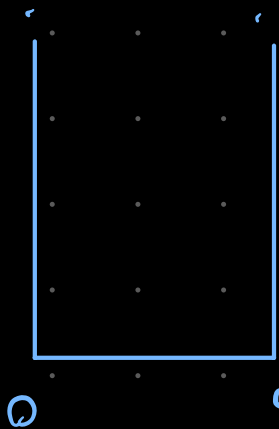
Square well simplification



Now Normalizable
 So for each Energy state there is only
 one possible stationary wave function

Solving the Infinite Square Well

$$V(x) = \begin{cases} \infty & x > a // x < 0 \\ 0 & a > x > 0 \end{cases}$$



$$\hat{H}\psi = E\psi$$

para $V(x) \rightarrow \infty$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \underset{\infty}{V(x)} \psi = E\psi$$

se $V(x) \uparrow \infty$ $\psi \rightarrow 0$

→ primeira condição (boundary condition)

se $V(x) = 0$ para $\underline{\psi = 0}$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi \rightarrow \frac{\partial^2 \psi}{\partial x^2} = -\frac{2m}{\hbar^2} E\psi$$

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2 \psi \rightarrow \text{clássico}$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$e^{ikx}$$

ou $A \sin(kx) + B \cos(kx)$

Por Didática

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

aplicando Boundary conditions

$$\psi(0) = 0 \quad \text{e} \quad \psi(a) = 0$$

$$A \sin(0) + B \cos(0) = 0$$

ou seja $B = 0$

resta $\psi(x) = A \sin(kx)$

$$A \sin(ak) = 0$$

Implica $ak = n\pi$

$$k = \frac{n\pi}{a}$$

$$k = \frac{\sqrt{2mE}}{\hbar} \quad \frac{\sqrt{2mE}}{\hbar} a = n\pi$$

$$\frac{2mE}{\hbar^2} a^2 = n^2 \pi^2 \Rightarrow E = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

Energia Quantizada

é possível perceber que a quantização
é emergente às limitações impostas

$$\psi(x) = A \sin\left(\frac{n\pi}{a}x\right)$$

Para normalizar a função de onda Estacionária no Espaço do Momento

Para Normalizar Probabilidade = 1

$$\int_0^a \psi^*(x) \psi(x) dx = 1$$

$$\int_0^a A^2 \sin^2\left(\frac{n\pi}{a}x\right) dx = 1$$

$$A^2 \int_0^a \sin^2\left(\frac{n\pi}{a}x\right) dx = 1$$

$$\underbrace{\int_0^a \sin^2\left(\frac{n\pi}{a}x\right) dx}_{= \frac{1}{2}(a-0)}$$

$$A^2 = \frac{2}{a}$$

$$A = \sqrt{\frac{2}{a}}$$

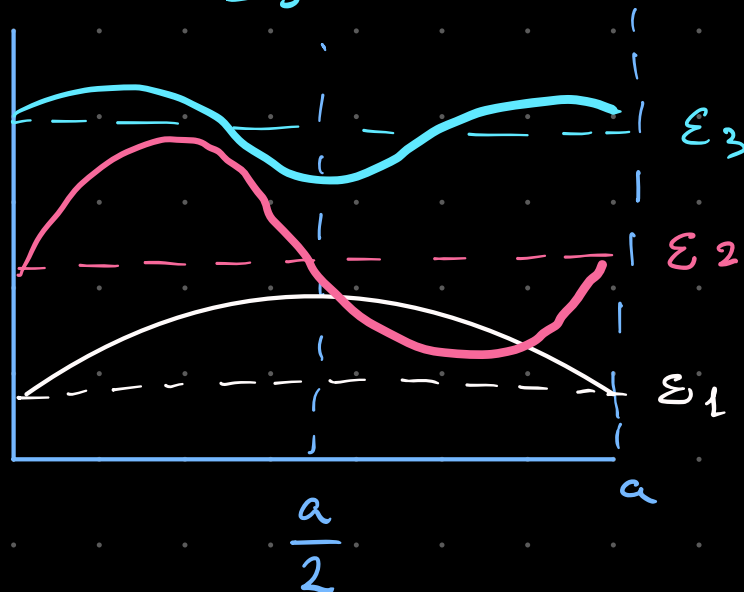
$$\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

Porte no Espaço do Momento de função de onda Estacionária

$$\Psi_s = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) e^{-i \frac{n^2 \pi^2 \hbar^2}{2ma^2} t}$$

função de onda estacionária

gráfico de $\Psi_s(x)$



Estados do ISW, orthogonality and completeness

Usando os resultados

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & x \geq a, x \leq 0 \end{cases}$$

$$\Psi_{s_n} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

Propriedades Gerais de Estados Estacionários

Symmetry $\rightarrow$ even or odd about middle of ISW

$n-1 \rightarrow$ number of nodes NOT in boundaries where $\psi \rightarrow 0$

Orthogonal

Complete basis

Ademais a ortogonalidade de função

vetores $\vec{A} \perp \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0$

$$a_x b_x + a_y b_y + a_z b_z = 0 \Rightarrow \sum_i a_i b_i = 0$$

$\hookrightarrow n$ dimensions

funções

$$\int f^*(x) g(x) dx \quad // \quad \int \psi^*(x) \psi(x) dx = 0$$

"like a dot product with itself"

$$\int f^*(x) g(x) dx = 0 \rightarrow \text{orthogonal}$$

Magnitude

$$\vec{a} \cdot \vec{a} = 1$$

unit vector

$$\int \psi^*(x) \psi(x) dx = 1$$

Normalization

Orthogonalidade de Seno

$$\int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx$$

Prosthema $\sin(x) \sin(y) = \frac{1}{2} [\cos(x-y) - \cos(x+y)]$

$$\frac{1}{2} \int_0^a \cos\left[\frac{n\pi x}{a} (n-m)\right] - \int_0^a \cos\left[\frac{n\pi x}{a} (n+m)\right]$$

$$\frac{1}{2} \frac{a}{(n-m)\pi} \sin\left[\frac{(n-m)\pi x}{a}\right] - \frac{a}{(n+m)\pi} \sin\left[\frac{(n+m)\pi x}{a}\right] \Big|_0^a$$

$$\frac{a\pi}{2} \left[\frac{\sin(n-m)\pi}{n-m} - \frac{\sin(n+m)\pi}{n+m} \right] = 0$$

$$\frac{\sin(n-m)\pi}{n-m} = \frac{\sin(n+m)\pi}{n+m}$$

δ_{nm} eh, o Kronecker delta

$$\int \psi^* \psi dx = \delta_{mn}$$

O Truque de Fourier

aplicado a ISW

Condição Inicial

$$f(x) = \sum_{n=1}^{\infty} C_n \underbrace{\sin\left(\frac{n\pi x}{a}\right)}_{\text{momentum space stationary states}}$$

momentum space stationary states

ψ_n

Como achar

C_n

$$\int_0^a \sin\left(\frac{m\pi x}{a}\right) f(x) dx = \int_0^a \sin\left(\frac{m\pi x}{a}\right) \sum_n C_n \sin\left(\frac{n\pi x}{a}\right) dx$$

$$\int_0^a \sin\left(\frac{m\pi x}{a}\right) f(x) dx = \sum_n C_n \underbrace{\int_0^a \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi x}{a}\right) dx}_{\substack{\text{Deu a ideia de ortogonalidade} \\ \text{do seno} \\ \text{essa expressão é 0} \\ \text{se } n \neq m \\ \int_n^m}}$$

$$\int_0^a \sin\left(\frac{m\pi x}{a}\right) f(x) dx = \sum_n C_n \frac{a}{2}$$

$$\frac{a}{2} C_m = \int_0^a \sin\left(\frac{m\pi x}{a}\right) f(x) dx$$

$$C_m = \frac{2}{a} \int_0^a \sin\left(\frac{m\pi x}{a}\right) f(x) dx$$

$$f(x) = \sum_n C_n \sin\left(\frac{n\pi x}{a}\right)$$

$$C_n = \frac{2}{a} \int_0^a \sin\left(\frac{n\pi x}{a}\right) f(x) dx$$

Vai ser revisa
nos

Exemplo $f(x) = \sin\left(\frac{3\pi}{a}x\right)$ Calcule C_1, C_2, C_3

$$\sin\left(\frac{3\pi}{a}x\right) = \sum C_n \sin\left(\frac{n\pi x}{a}\right)$$

$$\int \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{3\pi}{a}x\right) = \sum C_n \int \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{a}x\right)$$

$$\int_0^a \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{3\pi}{a}x\right) dx = C_n \frac{a}{2}$$

$$\frac{a}{2} C_n = \frac{a}{2} \quad C_n = 1 \quad C_1 = 1, C_2 = 1, C_3 = 1$$

Para $f(x) = -\frac{a}{2} + x$

$$C_m = \int_0^a \sin\left(\frac{m\pi}{a}x\right) \cdot \left[-\frac{a}{2} + x\right] dx$$

$$C_m = \int_0^a \sin\left(\frac{m\pi}{a}x\right) \cdot x \, dx - \int_0^a \sin\left(\frac{m\pi}{a}x\right) \frac{a}{2} dx$$

$$C_m = \frac{a \left(\sin\left(\frac{n\pi x}{a}\right) - \pi n \cos\left(\frac{\pi n x}{a}\right) \right)}{\pi^2 n^2} \Bigg|_0^a$$

$$+ \frac{a^2 \cos\left(\frac{\pi n x}{a}\right)}{2\pi n} \Bigg|_0^a$$

$$C_m = \frac{a^2 \left[\sin(n\pi) - \pi n \cos(n\pi) \right]}{\pi^2 n^2}$$

$$+ \frac{a^2 \cos(\pi n)}{2\pi n} - \frac{a^2}{2\pi n}$$

$$C_m = \frac{a^2 [-\pi n]}{\pi^2 n^2} + \frac{a^2}{2\pi n} - \frac{a^2}{2\pi n}$$

$$C_m = -\frac{a^2}{\pi n}$$

$$C_1 = -\frac{a^2}{\pi}$$

$$C_2 = -\frac{a^2}{2\pi}$$

$$C_3 = -\frac{a^2}{3\pi}$$

...

Resolvamos TISE e encontremos os estados estacionários

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right); \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

Momentum space stationary wave function

$$\Psi_n(x, \tau) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \cdot e^{-\frac{i n^2 \pi^2 \hbar}{2ma^2} \tau}$$

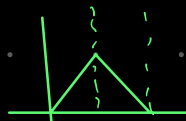
stationary wave function

Sendo

$$\Psi(x, 0) = \begin{cases} \sqrt{\frac{3}{a}} \left(1 - \left|\frac{x - \frac{a}{2}}{\frac{a}{2}}\right|\right) & 0 \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$$

conditioned Initials

$$\Psi(x, 0) = \begin{cases} \sqrt{\frac{3}{a}} \left(1 - \frac{x + \frac{a}{2}}{\frac{a}{2}}\right) & 0 < x < \frac{a}{2} \\ \sqrt{\frac{3}{a}} \left(1 + \frac{x - \frac{a}{2}}{\frac{a}{2}}\right) & \frac{a}{2} < x < a \\ 0 & \text{otherwise} \end{cases}$$



$$\Psi(x, 0) = \sum_n C_n \psi_n(x, 0)$$

$$C_n = \int_0^{\frac{a}{2}} \psi_n(x) \cdot \sqrt{\frac{3}{a}} \left(1 - \frac{x + \frac{a}{2}}{\frac{a}{2}}\right) dx$$

$$+ \int_{\frac{a}{2}}^a \psi_n(x) \cdot \sqrt{\frac{3}{a}} \left(1 + \frac{x - \frac{a}{2}}{\frac{a}{2}}\right) dx$$

$$C_n = \int_0^{\frac{a}{2}} \sin\left(\frac{n\pi x}{a}\right) \frac{\sqrt{3}}{\sqrt{4}} \left(1 - \frac{x + \frac{a}{2}}{\frac{a}{2}}\right) dx \\ + \int_{\frac{a}{2}}^a \sin\left(\frac{n\pi x}{a}\right) \frac{\sqrt{3}}{\sqrt{4}} \left(1 + \frac{x - \frac{a}{2}}{\frac{a}{2}}\right) dx$$

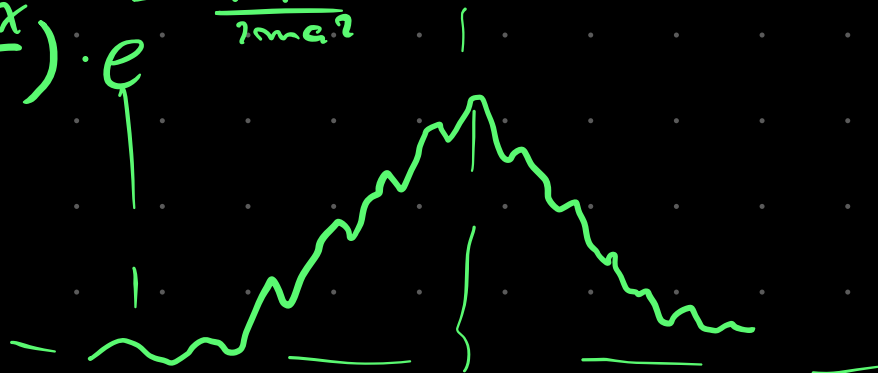
Jetzt wir sehen, dass es 41. resolved

$$C_n = \left[\frac{4\sqrt{2}\sqrt{3}}{\pi^2}, 0, \frac{4\sqrt{2}\sqrt{3}}{9\pi^2}, 0, \frac{4\sqrt{2}\sqrt{3}}{25\pi^2}, 0, \frac{-4\sqrt{2}\sqrt{3}}{49\pi^2} \right] \dots$$

$$\Psi(x, \tau) = \sum_n C_n \Psi_{\phi_n}(x, \tau)$$

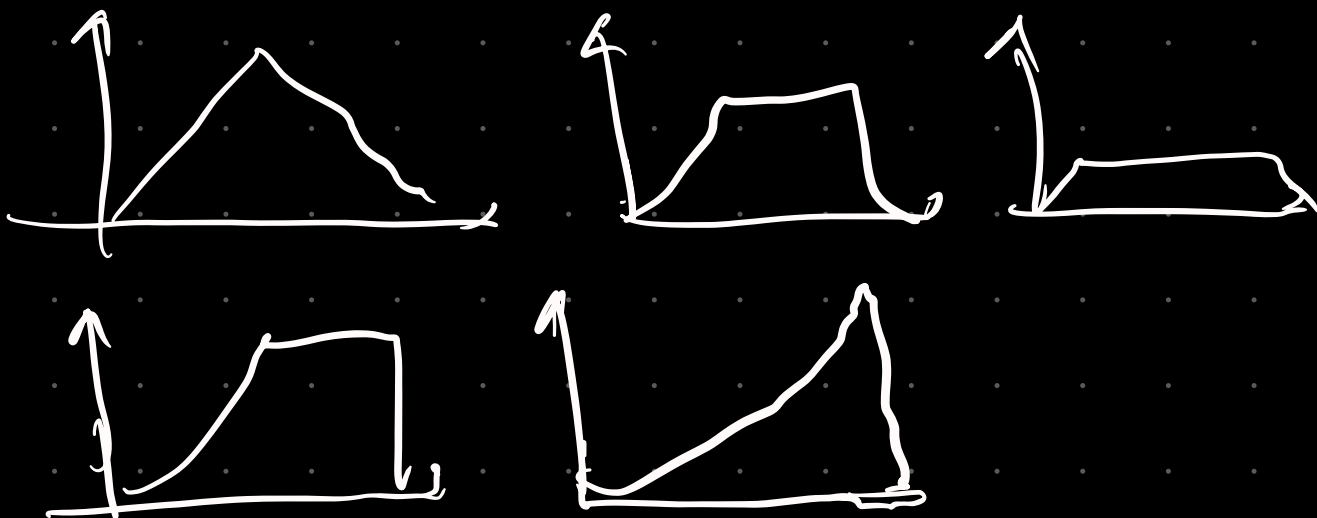
$$\Psi(x, \tau) = \frac{4\sqrt{2}\sqrt{3}}{\pi^2} \cdot \sin\left(\frac{\pi x}{a}\right) \cdot e^{-i \frac{\pi^2 \hbar}{2ma^2} \tau} + 0$$

$$\frac{-4\sqrt{2}\sqrt{3}}{9\pi^2} \cdot \sin\left(\frac{3\pi x}{a}\right) \cdot e^{-i \frac{9\pi^2 \hbar}{2ma^2} \tau}$$



also encounter $\langle x^2 \rangle =$

$$\langle x^2 \rangle = \int_0^a \Psi(x, \tau) \cdot x^2 \cdot \Psi^*(x, \tau)$$



Nota:

$$\Psi(x, \tau) = \sum_n C_n \psi(x) \cdot \psi(\tau)$$

"projeção média
do tempo"
de Fourier

↓
é o que muda nos estados
estacionários

$$\Psi(x, \tau) \cdot \psi^*(x) = \sum_n C_n \psi(x) \psi^*(x)$$

$$\int_0^a \psi^*(x) \cdot \Psi(x, \tau) = \int_0^a \sum_n C_n \psi(x) \psi^*(x)$$

estados estacionários são ortogonais a eles
mesmos

$$\int_0^a \psi^*(x) \Psi(x, \tau) = \sum_n C_n \int_0^a \psi^*(x) \psi(x)$$

$$\sum_n C_n \int_0^a \psi^*(x) \psi(x) = 1$$

$$\sum_n C_n \delta_n^m = C_m$$

Por $N = m$

uma variável

então

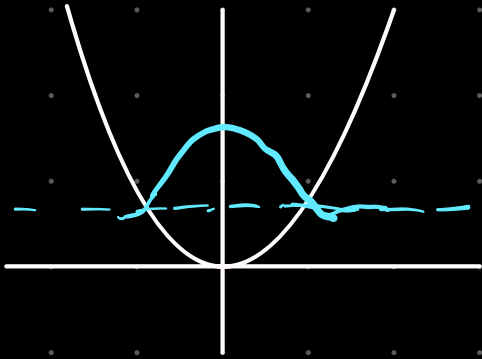
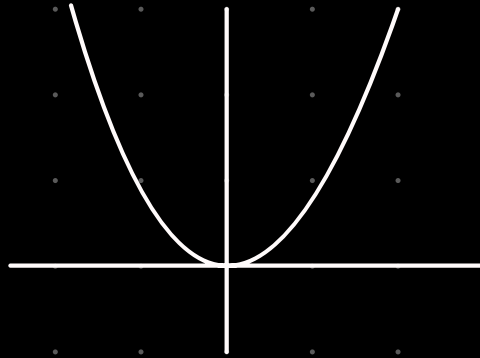
$$C_m = \int \psi^*(x) \cdot f(x) \cdot dx$$

Site legal

falstad.com | qml d

Quantum Harmonic Oscillator

$$V(x) = \frac{1}{2} k x^2$$



a frequência angular
de um oscilador
harmônico

$$\omega = \sqrt{\frac{k}{m}}$$

$$k = \omega^2 m$$

$$V(x) = \frac{\omega^2 m x^2}{2}$$

Serão apresentadas
duas soluções

a primeira com um truque com
operadores de escada ("ladder operators")

e a segunda na prática com "power series"
para resolver a EDO

Solution with ladder operators

começando com isto

$$V(x) = \frac{m\omega^2 x^2}{2} \quad \hat{K}\psi = \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} \quad \hat{V}\psi = \frac{m\omega^2 x^2}{2} \psi$$

$$\hat{H}\psi = \epsilon\psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \frac{m\omega^2 x^2}{2} \psi = \epsilon\psi$$

→ Sabendo que $\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{m\omega^2 x^2}{2}$

|| fazendo essa expressão

$$(a+b)(a-b) = (a-b)^2$$

$$\frac{1}{2m} \left(m^2 \omega^2 x^2 - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right)$$

lembrando que operações $\frac{1}{2m} ((m\omega x)^2 - \hat{p}^2)$
e produtos matriciais com operadores não
são necessariamente válida

$$\frac{1}{2m} (m\omega \hat{x} - \hat{p})(m\omega \hat{x} + \hat{p}) \neq \frac{1}{2m} ((m\omega x)^2 - \hat{p}^2)$$

$$\frac{1}{2m} (m^2 \omega^2 \hat{x}^2 + m\omega \hat{x} \hat{p} - \hat{p} \hat{x} m\omega - \hat{p}^2)$$

Commutator: $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$

$$\frac{1}{2m} (m^2 \omega^2 \hat{x}^2 + m\omega (\hat{x}\hat{p} - \hat{p}\hat{x}) - \hat{p}^2)$$

$$\frac{1}{2m} (m^2 \omega^2 \hat{x}^2 + m\omega [\hat{x}, \hat{p}] - \hat{p}^2)$$

Perceba que a que difere a expressão original desta Imagem é

$$\frac{\omega}{2} [\hat{x}, \hat{p}]$$

$$\hat{x} \left(-i\hbar \frac{\partial}{\partial x} \right) - \left(-i\hbar \frac{\partial}{\partial x} \right) \hat{x}$$

$$- \hat{x} i\hbar \frac{\partial \psi}{\partial x} + i\hbar \frac{\partial \hat{x} \psi}{\partial x}$$

$$- i\hbar \hat{x} \frac{\partial \psi}{\partial x} + i\hbar \left(\hat{x} \frac{\partial \psi}{\partial x} + \psi \frac{\partial \hat{x}}{\partial x} \right)$$

$$i\hbar \left[\hat{x} \frac{\partial \psi}{\partial x} - \frac{\partial \hat{x}}{\partial x} \psi + \psi \frac{\partial \hat{x}}{\partial x} \right]$$

$$[\hat{x}, \hat{p}] \psi = i\hbar \psi \frac{\partial}{\partial x} = \underbrace{i\hbar}_{\text{equiv.}} \psi$$

equiv. as from experience é $\frac{i\hbar \omega}{2}$

$$\frac{1}{2m} (m^2 \omega^2 \hat{x}^2 - \hat{p}^2) + \frac{i\hbar \omega}{2} = \frac{1}{2m} (m\omega \hat{x} - \hat{p})(m\omega \hat{x} + \hat{p})$$

Ladder operators

$$\hat{A}_{\pm} = \sqrt{\frac{1}{2\pi m\omega}} (\pm i\hat{p} + m\omega \hat{x})$$

$$\hat{H} = \hat{A}_+ \hat{A}_- \rightarrow \text{for } \text{harmonic} \text{ é } \text{add}$$

$$\hat{H} = \frac{1}{2\pi m\omega} (i\hat{p} + m\omega \hat{x})(m\omega \hat{x} - i\hat{p})$$

$$\hat{H} = \hat{p}^2 + m^2 \omega^2 \hat{x}^2 + [\hat{x}, \hat{p}]$$

$$\hat{H}\psi = \epsilon \psi$$

$$\hbar \omega \left(\hat{A}_+ \hat{A}_- + \frac{1}{2} \right) \psi = \epsilon \psi$$

$$\hbar\omega(\hat{a}_+\hat{a}_- + \frac{1}{2})\psi = \epsilon\psi$$

Para Dirac

considero que

$$\hat{H} \hat{a}_+\psi$$

$$\hbar\omega(\hat{a}_+\hat{a}_- + \frac{1}{2})\hat{a}_+\psi = \epsilon\psi$$

$$\hbar\omega(\hat{a}_+\hat{a}_-\hat{a}_+ + \frac{1}{2}\hat{a}_+)\psi = \epsilon\psi$$

$$\hat{a}_+\hbar\omega(\hat{a}_-\hat{a}_+ + \frac{1}{2})\psi = \epsilon\psi$$

$$\hbar\omega\hat{a}_+(\hat{a}_-\hat{a}_+ - \frac{1}{2} + 1)\psi = \epsilon\psi$$

$$\hbar\omega\hat{a}_+\hat{a}_-\hat{a}_+ - \frac{\hbar\omega}{2}\hat{a}_+ + \hbar\omega\hat{a}_+$$

$$\underbrace{(\hbar\omega\hat{a}_+\hat{a}_- - \frac{\hbar\omega}{2})\hat{a}_+ + \hbar\omega\hat{a}_+}_{\hat{H}}$$

$$\hat{a}_+(\hat{H} + \hbar\omega)\psi = \epsilon\psi$$

$$\hat{a}_+(\epsilon + \hbar\omega)\psi = \epsilon\psi$$

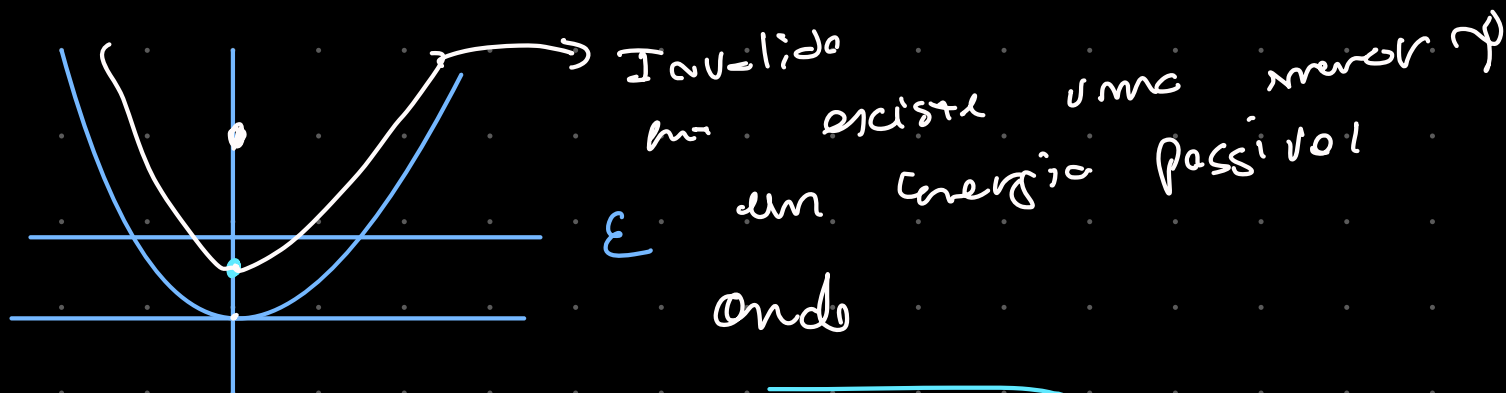
$$\hat{H}(\hat{a}_+ \psi) = (\epsilon + \hbar\omega) \hat{a}_+ \psi$$

"Novo E_0 Sch"

se ψ é sol $\hat{a}_+ \psi$ tbm é solução

Ladder Operators and Ground State

$$\psi_{n-2} \xrightarrow{\hat{a}_+} \psi_{n-1} \xrightarrow{\hat{a}_+} \psi_n \xrightarrow{\hat{a}_+} \psi_{n+1} \xrightarrow{\hat{a}_+} \psi_{n+2} \xrightarrow{\hat{a}_+} \psi_{n+3}$$



$$\hat{a}_- \psi_0 = 0$$

$$\frac{1}{\sqrt{2m\hbar\omega}} \left(\hbar \frac{d}{dx} + m\omega x \right) \psi_0 = 0$$

$$\hbar \frac{d\psi_0}{dx} + m\omega x \psi_0 = 0$$

$$\hbar \frac{d\psi_0}{dx} = -m\omega x \psi_0$$

$$\frac{d\psi_0}{dx} = -\frac{m\omega x \psi_0}{\hbar}$$

$$\psi_0 = A e^{-\frac{m\omega x^2}{2\hbar}}$$

Normalisieren so

$$1 = \int_{-\infty}^{\infty} \psi_0^* \psi_0 dx$$

$$1 = \int_{-\infty}^{\infty} A e^{-\frac{m\omega x^2}{\hbar}} dx$$

Solution by Power Series

$$\hat{H}\psi = E\psi \quad \hat{H} = \frac{\hat{p}^2}{2m} - m\omega^2 \hat{x}^2$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - m\omega^2 \hat{x}^2 \psi = E\psi$$

$$(E + m\omega^2 \hat{x}^2) \psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

$$-\frac{2m}{\hbar^2} (E + m\omega^2 \hat{x}^2) \psi = \frac{\partial^2 \psi}{\partial x^2}$$

$$\left(\frac{2m^2 \omega^2 x^2}{\hbar^2} - \frac{2mE}{\hbar^2} \right) \psi = \frac{\partial^2 \psi}{\partial x^2}$$

$$(\xi^2 - K) \psi = \frac{\partial^2 \psi}{\partial x^2}$$

$$\xi = \sqrt{\frac{2m^2 \omega^2}{\hbar^2}} x$$

$$\frac{m\omega}{\hbar} \sqrt{2x^2}$$

$$K = \frac{2mE}{\hbar^2}$$

A particle lives

$$\hat{H}\psi = E\psi \Rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

a solution is possible even trivial

$$-\frac{2mE}{\hbar^2} \psi = \frac{\partial^2 \psi}{\partial x^2} \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$-k^2 \psi = \frac{\partial^2 \psi}{\partial x^2} \xrightarrow{E>0} A e^{ikx} + B e^{-ikx} = \psi(x)$$

$$\Psi(x, \tau) = A e^{ikx} e^{-i\frac{E\tau}{\hbar}} + B e^{-ikx} e^{-i\frac{E\tau}{\hbar}}$$

$$\Psi(x, \tau) = A e^{i(kx - \frac{E\tau}{\hbar})} + B e^{-i(kx + \frac{E\tau}{\hbar})}$$

$$E = \frac{\hbar^2 k^2}{2m} \quad A e^{ik(x - \frac{E\tau}{\hbar})} + B e^{-ik(x + \frac{E\tau}{\hbar})}$$

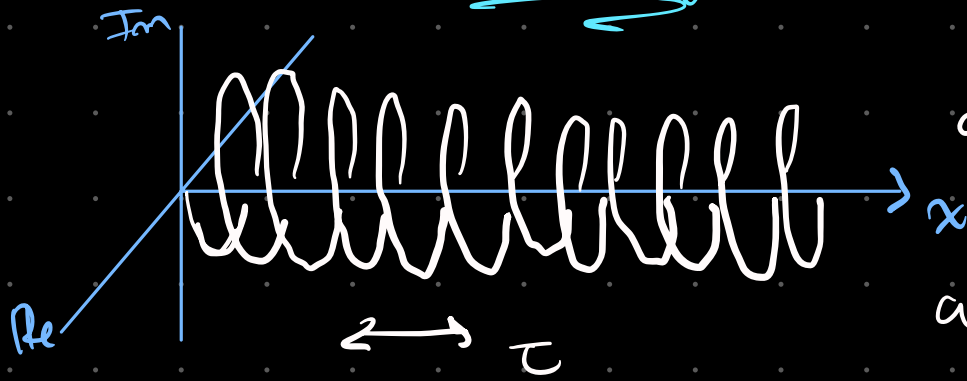
$$\Psi(x, \tau) = A e^{ik(x - \frac{E\tau}{\hbar})} + B e^{-ik(x + \frac{E\tau}{\hbar})}$$

its clear from $\langle x \rangle \rightarrow \langle \Psi | x | \Psi \rangle$

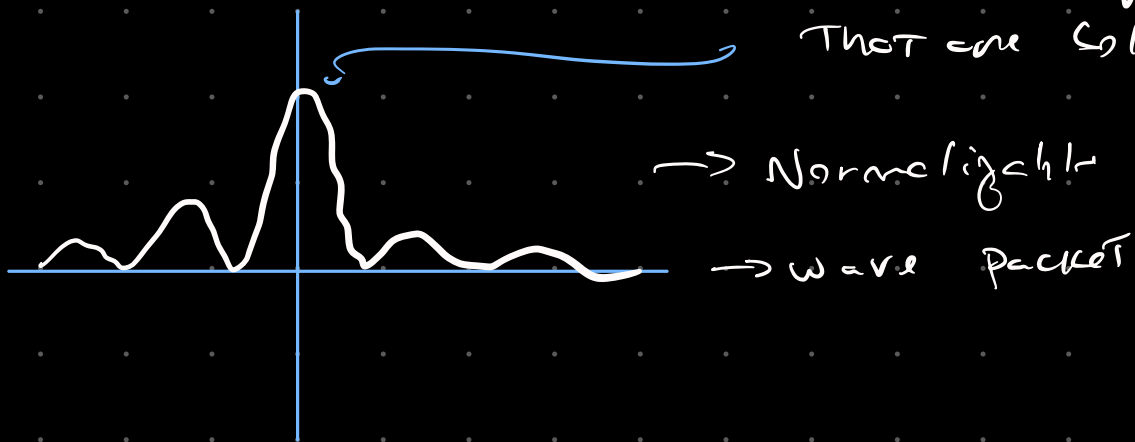
that this is a travelling wave

because both terms $A, B \neq 0$

but as a direct consequence of this
 There are no Boundary Conditions
 and so Normalization is impossible and
 it is not Quantized



as the function
 is not normalizable
 a particle must
 be a superposition
 of travelling waves
 that are solutions



The Wave Packet and Stationary States

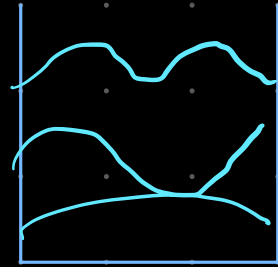
Properties of the wave packet

$\psi = 0$ for large $|x|$ condition and non zero for small

Building the Wave Packet

for the Infinite square well we had

$$\Psi(x,t) = \sum_{n=1}^{\infty} C_n \underbrace{\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)}_{\text{a stationary state}} e^{-i \frac{n^2 \pi^2 \hbar}{2ma^2} t}$$



where in the ISW thanks to the boundary conditions we had a sum of quantized stationary states

In the case of the free particle the function is continuous and not quantized

↪ $e^{ik(x - \frac{\hbar k^2}{2m}t)}$ where k is an unrestricted parameter

$$\sum C_n e^{ikx}$$

↪
 $\int dk$

↪ Don't make sense, as to be a function of k it needs to be a continuous summation given by $\phi(k)$ instead of C_n

$$\Psi(x,t) = \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m}t)} dk$$

↪ how to find $\phi(k)$

Revisiting Fourier's Trick for analogy

$$\psi(x,0) = \sum c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}\right)x$$

$$\int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi}{a}\right)x \psi(x,0) dx = \sum_{n=1}^{\infty} c_n \frac{2}{a} \int_0^a \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{a}\right)x dx$$

$$c_m = \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi}{a}\right)x \psi(x,0) dx$$

$$\psi(x,0) = \int_{-\infty}^{\infty} \phi(k) e^{ikx} dk$$

$$\int_{-\infty}^{\infty} e^{-ik'x} \psi(x,0) dx =$$

$$\int_{-\infty}^{\infty} dk \phi(k) \int_{-\infty}^{\infty} dx e^{-ik'x} e^{ikx}$$

$$= \int_{-\infty}^{\infty} dk \phi(k) \delta(k-k')$$

$$= \phi(k')$$

we need an orthogonality condition

$$\delta(k-k')$$

a dirac delta function

$$k=k'=L$$

$$L \neq k' = 0$$

$$\phi(k') = \int_{-\infty}^{\infty} e^{-ik'x} \psi(x,0) dx$$

fourier analysis

for our case of the Free Particle.

$$\Psi(x, z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m} z)} dk$$

where $\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx$

Example find $\phi(k)$ for $\Psi(x, 0) = \begin{cases} \frac{1}{\sqrt{2a}} & -a \leq x \leq a \\ 0, & \text{otherwise} \end{cases}$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2a}} e^{-ikx} dx$$

$$= \frac{1}{2\sqrt{\pi a}} \int_{-\infty}^{\infty} e^{-ikx} dx$$

$$\phi(k) = -\frac{1}{2\sqrt{\pi a}} \cdot \frac{1}{ik} e^{-ikx} \Big|_{-\infty}^{\infty}$$

Wave Velocity

$$\Psi(x, z) = e^{i(kx - \frac{\hbar k^2}{2m} z)}$$

$$kx - \frac{\hbar k^2}{2m} z = 0$$

locus of +ve

$$\frac{x}{z} = \frac{\hbar k}{2m}$$

$$kx \uparrow = \frac{\hbar k^2}{2m} z \uparrow$$

moves in $\oplus$

$$V = \frac{\hbar k}{2m}$$

$$V = \frac{\hbar}{2m} \sqrt{\frac{2mE}{\hbar^2}} \Rightarrow$$

$$V = \sqrt{\frac{E}{2m}}$$

velocity
of +

classically $\frac{1}{2}mv^2 = E \Rightarrow v = \sqrt{\frac{2E}{m}}$ a feature of the wave

Consider the wave packet

$$\Psi(x,t) = e^{i(k_1 x - \frac{\hbar k_1^2}{2m} t)} + e^{i(k_2 x - \frac{\hbar k_2^2}{2m} t)}$$

This is not actually a wave packet

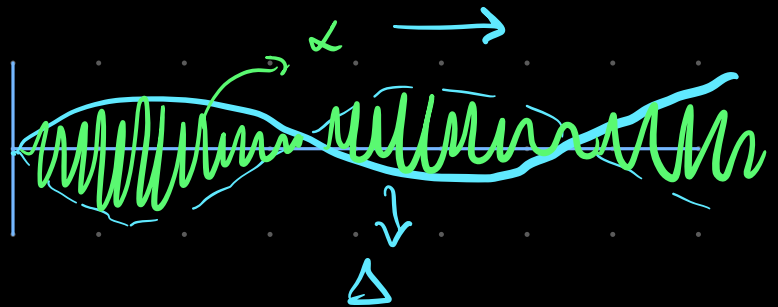
$$\alpha = \frac{k_1 + k_2}{2} x - \frac{\hbar(k_1^2 + k_2^2)}{4m} t \rightarrow \text{Big}$$

$$\Delta = \frac{k_1 - k_2}{2} x - \frac{\hbar(k_1^2 - k_2^2)}{4m} t \rightarrow \text{small}$$

$$\Psi(x,t) = e^{i(\alpha + \Delta)} + e^{i(\alpha - \Delta)}$$

$$\Psi(x,t) = e^{i\alpha} (e^{i\Delta} + e^{-i\Delta})$$

$$= e^{i\alpha} 2\cos(\Delta)$$



The velocity of the wave depends on the slowly varying envelope Δ

$$\Delta = 0 \Rightarrow \frac{k_1 - k_2}{2} x = \frac{\hbar}{4m} (k_1 + k_2) (k_1 - k_2) t$$

$$\frac{x}{t} = \frac{\hbar}{2m} \underbrace{(k_1 + k_2)}_{2k}$$

$$v = \frac{\hbar \vec{k}}{m} = \frac{\hbar}{m} \frac{\sqrt{2mE}}{\hbar} \Rightarrow v \approx \sqrt{\frac{2E}{m}}$$

ITS slower than Classical

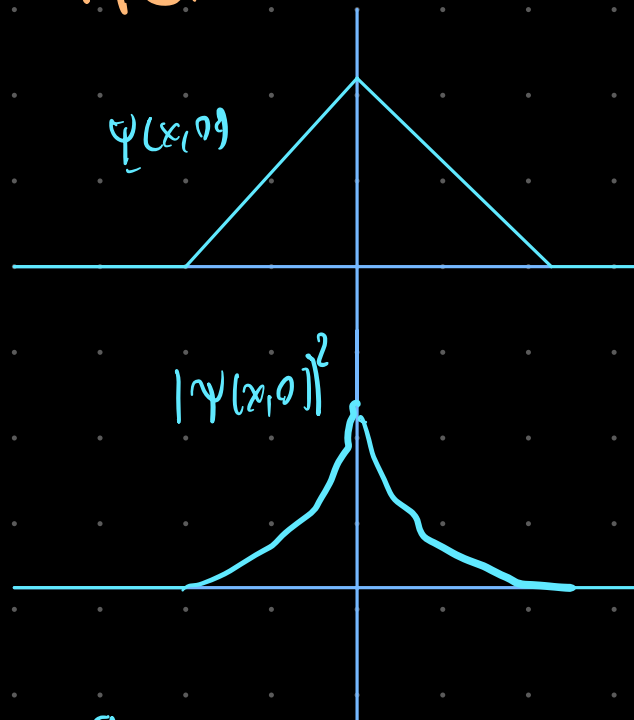
group velocity and phase velocity

BUT with correct energy

Example of Free Particle

Consider the initial conditions

$$\psi(x,0) = \sqrt{\frac{3}{2a}} \begin{cases} \frac{a+x}{a} & -a \leq x < 0 \\ \frac{a-x}{a} & 0 \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$$



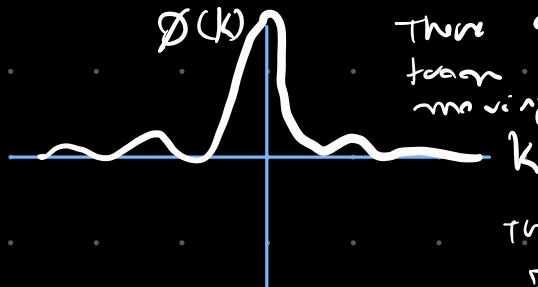
$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x,0) e^{-ikx} dx$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{3}{2a}} \left[\int_{-a}^0 \frac{a+x}{a} e^{-ikx} dx + \int_0^a \frac{a-x}{a} e^{-ikx} dx \right]$$

$\text{Sinc}^2(x)$

using integration by parts

$$\phi(k) \sim \frac{\text{Sinc}^2(x)}{x^2}$$



There are equal contributions from right moving and left moving wave so

It's expected the two fine lines doesn't move

$$\Psi(x, \tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m} \tau)} dk$$

$$\Psi(x, \tau) = \frac{L}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(k)}{k^2} e^{i(kx - \frac{\hbar k^2}{2m} \tau)} dk$$

Solving this you can see a wave packet that delocalizes with time